Three Important Medium in Cordyceps Militaris

PDA MEDIUM (POTATO DEXTROSE AGAR MEDIUM)		LIQUID MEDIUM (LIQUID SPAWN / COMMERICAL SPAWN)		BASAL MEDIUM	
Particulars	For 1 Litre Preparation	Particulars	For 1 Litre Preparation	Particulars	For 1 Litre Preparation
Potato Broth (Juice)	1 Liter (1000 m.l)	Potato Broth (Juice)	1 Liter (1000 m.l)	Potato Broth (Juice)	1 Liter (1000 m.l)
Agar	20 grams	----------	------	---------	------
Mg So4	500 m. grams	Mg So4	500 m. grams	Mg So4	500 m. grams
Dextrose	20 grams	Dextrose	30 grams	Dextrose	30 grams
		Peptone	10 grams	Peptone	10 grams
--------	-------	Yeast	1 gram	Yeast	1 gram
--------	-------	Vitamin – B 1	100 m. grams	Vitamin – B 1	100 m. grams
--------	-------	Vitamin – B 12	100 m. grams	Vitamin – B 12	100 m. grams
--------	-------	--------	--------	Egg (Not compulsory for Vegetarian)	01

Agar = A Gelatinous Substance that is Extracted from Seaweed (கடற்பாசி)
Dextrose = Glucose Mg So4 = Magnesium Sulphate

Preparation of Potato Broth (Juice)

Required Materials :- Potato & R.O Water (Best :- Sterile R. O Water)

Material	Measurements											
Required Potato Broth	100 m.l	150 m.l	200 m.l	250 m.l	300 m.l	400 m.l	500 m.l	1 Liter	1.5 Liters	2.0 Liters	2.5 Liters	3.0 Liters
Potato	20 gms.	30 gms.	40 gms.	50 gms.	60 gms.	80 gms.	100 gms.	200 gms.	300 gms.	400 gms.	500 gms.	600 gms.
R. O Water	100 m.l	150 m.l	200 m.l	250 m.l	300 m.l	400 m.l	500 m.l	1 Liter	1.5 Liters	2.0 Liters	2.5 Liters	3.0 Liters

Preparation of Potato Broth for 3 Medium (PDA Medium + Liquid Medium + Basal Medium)

For Preparation required Potatoes

Actual Required Potatoes + Extra Potatoes

(for Skin Wastage + Unwanted damaged Potatoes like wrinkled skins, Cut Surface, soft dark areas, with Green in Appearance, badly Sprouted and Bruises)

For Preparation required R.O Water

Actual Required R.O. Water + Extra R.O Water

(Water boiling evaporation + Additional)

	PDA MEDIUM (POTATO DEXTROSE AGAR)					LIQUID MEDIUM (LIQUID SPAWN / COMMERICAL SPAWN)				BASAL MEDIUM	
Particulars	For 100 m.l	For 150 m.l	For 200 m.l	For 250 m.l	For 300 m.l	For 1 Liter	For 2 Liters	For 3 Liters	For 4 Liters	For 1 Liter	For 2 Liters
Potato Broth (Juice)	100 m.l	150 m.l	200 m.l	250 m.l	300 m.l	1 Liter	2 Liters	3 Liters	4 Liters	1 Liter	2 Liters
Agar	2 grams	3 grams	4 grams	5 grams	6 grams	—	—	—	—	—	—
Mg So4	50 m. grams	75 m. grams	100 m. grams	125 m. grams	150 m. grams	500 m. grams	1.0 gram	1.5 grams	2.0 grams	500 m. grams	1.0 gram
Dextrose	2 grams	3 grams	4 grams	5 grams	6 grams	30 grams	60 grams	90 grams	120 grams	30 grams	60 grams
Peptone	→	→	→	→	→	10 grams	20 grams	30 grams	40 grams	10 grams	20 grams
Yeast	→	→	→	→	→	1 gram	2 grams	3 grams	4 grams	1 gram	2 grams
Vitamin B-1	→	→	→	→	→	100 m.grams	200 m.grams	300 m.grams	400 m.grams	100 m.grams	200 m.grams
Vitamin B-12	→	→	→	→	→	100 m.grams	200 m.grams	300 m.grams	400 m.grams	100 m.grams	200 m.grams
Egg	Not compulsory for Vegetarian					→	→	→	→	01	02

Whisky Bottle {or} Test Tube
35 m.l to 40 m.l — 10 m.l to 15 m.l

Conical Flask {or} Saline Bottle
250 m.l to 300 m.l

Cultivation Cup/ Lug Jar
For 1 gram rice 1.6 m.l Basal Medium

How to Prepare Potato Broth (Juice)

Clean the required Potatoes half-hour before, with ordinary water

Take required Potatoes and weigh them

Remove all the Potato skin with help of a Peeler

Take the Peeled Potatoes and cut into small pieces
After cutting Potatoes, immediately put them in R.O water. This will help preventing the Potatoes from browning.

Collect all the cut Potatoes without R.O water and drop them into a big Stainless Steel Vessel.
Pour measured Sterile R.O water (or) if Sterile water is not available pour measured fresh R.O Water
Cover the SS Vessel with lid

Switch ON the Stove (with the help of Gas Stove or Induction Stove)
Boil or Cook the Potatoes for 15 Minutes, till the Potatoes become soft condition. Now Switch Off the Stove
Leave it for 5 minutes

Open the Lid. Now drain and collect the Potato Juice alone in other Stainless Steel Vessel with the help of a Sieve
Now the Potato Broth is ready

Peeler -- a device or tool for removing the skin from fruit and vegetables.
Sterile: Free from bacteria or other living microorganisms; totally clean. R.O: Reverse Osmosis

How to Prepare Potato Broth (Juice)

Clean the required potatoes half-hour before, with ordinary water	**தேவையான உருளைக்கிழங்கை அரை மணி நேரத்திற்கு முன், சாதாரண தண்ணீரில் சுத்தம் செய்யவும்**
Take required potatoes and weigh them	தேவையான உருளைக்கிழங்கை எடுத்து எடை போடவும்
Remove the potato skin with help of Peeler	பீலர் உதவியுடன் உருளைக்கிழங்கு தோலை அகற்றவும்
Take the Peeled Potatoes and cut into small pieces	தோலுரித்த உருளைக்கிழங்கை எடுத்து சிறிய துண்டுகளாக வெட்டவும்
After cutting Potatoes, immediately put them in R. O water. This will help preventing the potatoes from browning.	உருளைக்கிழங்கை வெட்டிய உடனே ஆர்.ஓ தண்ணீரில் போடவும். இது உருளைக்கிழங்கு பழுப்பு நிறமாவதைத் தடுக்க உதவும்.
Collect all the cut potatoes without R.O water and drop them into a big Stainless Steel Vessel. Pour measured Sterile R.O water (or) if Sterile water is not available pour measured fresh R.O Water Cover the SS Vessel with lid	ஆர் ஓ தண்ணீர் இல்லாமல் வெட்டப்பட்ட அனைத்து உருளைக்கிழங்குகளையும் சேகரித்து, அவற்றை ஒரு பெரிய துருப்பிடிக்காத எஃகு பாத்திரத்தில் விடவும். அளவிடப்பட்ட ஸ்டெரைல் ஆர் ஓ தண்ணீரை ஊற்றவும் (அல்லது) ஸ்டெரைல் ஆர் ஓ தண்ணீர் கிடைக்கவில்லை என்றால் புதிய அளவிடப்பட்ட ஆர் ஓ தண்ணீரை ஊற்றவும் SS பாத்திரத்தை மூடியால் மூடி வைக்கவும்
Switch ON the Stove (with the help of Gas Stove or Induction Stove) Boil or Cook the Potatoes for 15 Minutes, till the Potatoes become soft condition. Now Switch Off the Stove Leave it for 5 minutes	அடுப்பை இயக்கவும் (கேஸ் ஸ்டவ் அல்லது இண்டக்ஷன் ஸ்டவ் உதவியுடன்) உருளைக்கிழங்கை 15 நிமிடங்களுக்கு வேகவைக்கவும் அல்லது சமைக்கவும், உருளைக்கிழங்கு மென்மையாக மாறும் வரை. இப்போது அடுப்பை அணைக்கவும் 5 நிமிடங்கள் அப்படியே விடவும்
Open the Lid. Now drain and collect the Potato Juice alone in other Stainless steel Vessel with the help of a Sieve Now the Potato Broth is ready	மூடியைத் திறக்கவும். இப்போது உருளைக்கிழங்கு சாற்றை தனியாக வடிகட்டி மற்ற எஃகு பாத்திரத்தில் ஒரு சல்லடையின் உதவியுடன் சேகரிக்கவும். இப்போது உருளைக்கிழங்கு குழம்பு தயார்
Peeler -- a device or tool for removing the skin from fruit and vegetables. Sterile: Free from bacteria or other living microorganisms; totally clean. **R.O: Reverse Osmosis**	**பீலர் -- பழங்கள் மற்றும் காய்கறிகளிலிருந்து தோலை அகற்றுவதற்கான ஒரு சாதனம் அல்லது கருவி.** நோய் நுண்மங்கள் ஒழிக்கப்பெற்ற: பாக்டீரியா அல்லது பிற வாழும் நுண்ணுயிரிகளிலிருந்து பிரிக்கப்பட்ட முற்றிலும் தூய்மையானது ஆர்.ஓ: தலைகீழ் சவ்வூடுபரவல்

CORDYCEPS MILITARIS

Culture Section

Base Culture

PDA MEDIUM

{Potato + Dextrose + Agar}

[Medium has to be prepared as per the provide chart]

STERILIZE

After preparation of PDA Medium, distribute immediate in the Whisky Bottle {30 to 40 m.l} (or) Boiling Tube {10 to 15 m.l} Before it change to Solid form. Cover with Absorbent Cotton and Cut or News Paper.

Keep in the Autoclave at 15 PSI / 121°C for 40 Minutes

WAITING PERIOD

After 40 minutes Sterilization take out the PDA Medium from the Autoclave. Keep all the PDA Medium bottle In Slanting [10° to 15°] Position for 5 days

Discard any Contaminated PDA Medium is seen during the waiting period 5 days

ISOLATE & INOCULATE

Cut a slice from the top of the well – grown Mother Mushroom and place it at Centre of the PDA Medium

[Whisky Bottle or Boiling tube, After 5 Days waiting Period]

Temperature – 20° C to 22° C

Dried Mushroom – 20 % Success

Fresh Mushroom – 80 % Success

INCUBATE

Keep in Dark + 20 Days + Temperature 22°C

[Dark – Cover with Black or Blue Thick Cloth]

Remove the Cloth Cover before 2 days. (19th day)

Give diffused light on 19th day & 20th day. Now the Culture changes from White to Yellow and Orange Colour.

NOW THE BASE CULTURE IS READY

Liquid Culture

LIQUID MEDIUM

[Medium has to prepared as per the provide chart]

STERILIZE

After preparation of Liquid Medium distribute in the Saline bottle (or) Conical Flask (250 to 300 m.l)

Cover with Non Absorbent Cotton and Paper

Keep in the Autoclave at 15 PSI / 121°C for 40 Minutes

WAITING PERIOD

After 40 minutes Sterilization take out the Liquid Medium from the Autoclave. Keep all the Liquid Medium in Normal Position for cooling 1 day only (less than 24 Hrs)

ISOLATE & INOCULATE

(With 20 Days Old Base Culture)

Cut a 1 C.M Square Piece from the 20 days Old Solid Base Culture and put inside the Sterilized Liquid Medium

Temperature 20°C to 22°C

INCUBATE

Keep in Dark + 20 Days + Temperature 22°C

[Dark – Cover with Black (or) Blue Colour Thick Cloth]

{ If We need More Conidia – From 2nd day Onwards }

Rotate the Saline Bottles

5 times Clockwise and 5 times Anti Clockwise

[Daily Morning and Evening for 5 days only, up to 6th Day]

Let Stand Static for the remaining 14 Days

If We don't need more Conidia - Let Stand Static all the days

Remove the Cloth Cover before 2 days (19th Day)

Give diffused light 19th & 20th day. Now the Culture Changes from White to Yellow and then Orange Colour

NOW THE LIQUID (OR) COMMERCIAL SPAWN IS READY

Cropping Section

BASAL MEDIUM

[Medium has to be prepared as per the provide chart]

SUBSTRATE

Take 20 grams Brown Rice in Lug Jar (or) Glass Bottle and pour (mix) 32 m.l. Basal Medium

[For 1 gram Rice 1.6 m.l Basal Medium is required]

STERILIZE & WAITING PERIOD

After preparation of the Lug Jar (or) Glass Bottle, cover the Lid Air hole with absorbent Cotton (or) Micropore tape

Keep in the Autoclave at 15 PSI 121°C for 40 minutes

After 40 minutes Sterilization take out the Lug Jar/ Bottle from the Auto clave. Keep all the Bottle in Normal Position for cooling 1 day (less than 24 Hrs.)

INOCULATE & INCUBATE

Take Sterilize Lug Jar / Glass Bottle

Add (pour) 10 m.l to 15 m.l Liquid Spawn

Keep in Dark + Temperature 22°C + 10 Days

[Dark – Cover with Black / Blue Colour Thick Cloth]

STRESS / SHOCK

Remove the Cloth Cover over the Lug jar/ Glass Bottle on 11th day

Give 500 Lux diffused light from 11th day onwards

Give 700 Lux diffused light from 16th day onwards from 21st day onwards up to 60th day

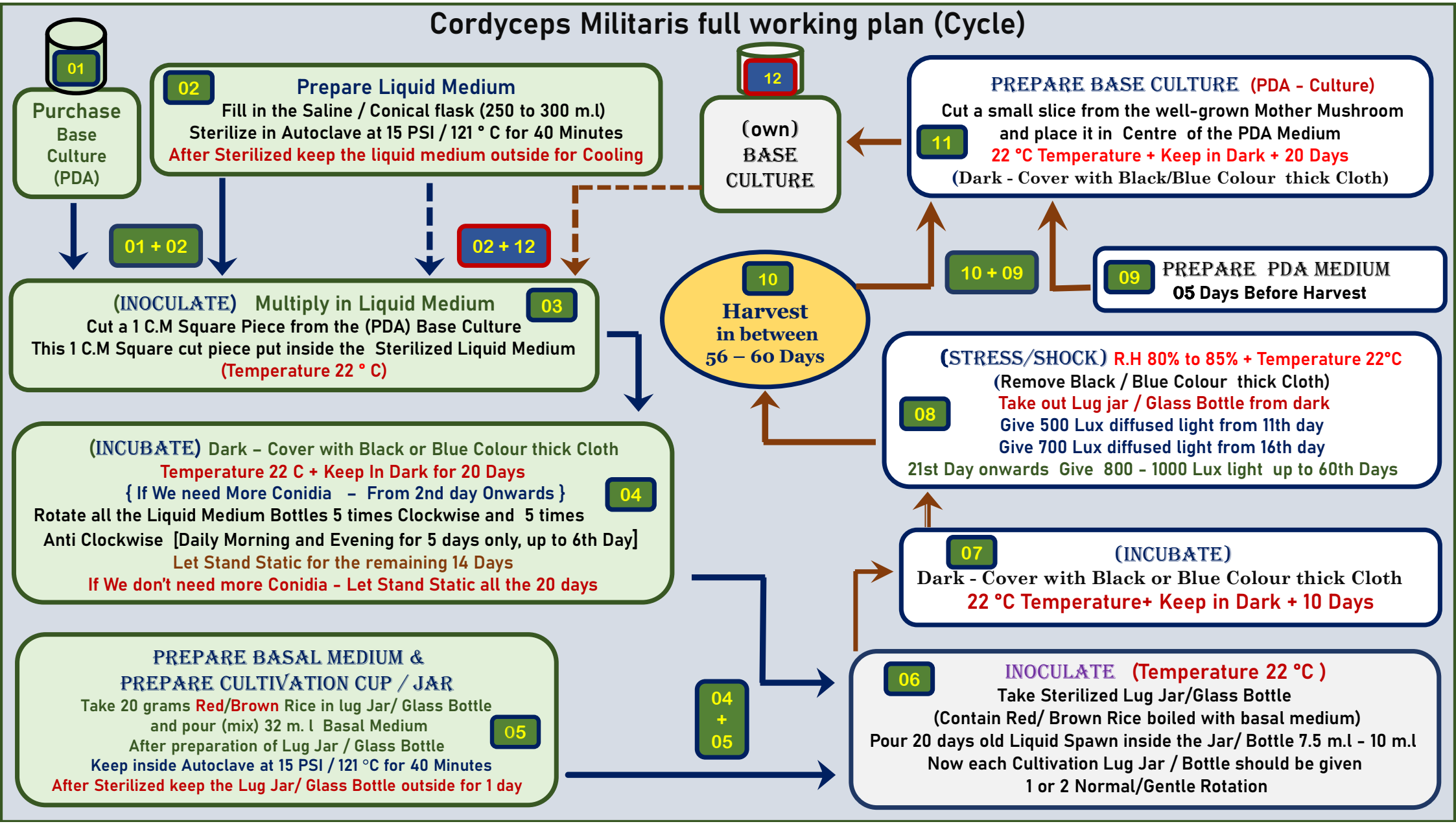
Give 800 to 1000 Lux direct Light

Temperature 22°C + R.H 80 % to 85 %

CROP HARVEST

Harvest can be done between 56th Day to 60th Day

Cordyceps Militaris full working plan (Cycle)

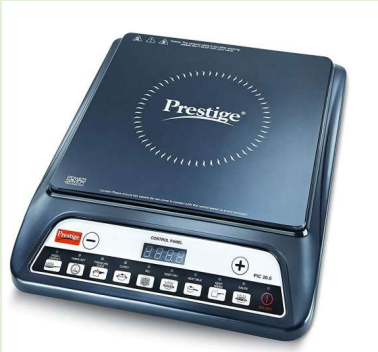


NECESSARY ITEMS {Instruments, Apparatus & Minimum tools} FOR CORDYCEPS MILITARIS

Bottle cooler | **Autoclave (Manual)** | **Weighing Machine (Option)** | **Precision Balance** | **Single Gas Burner**

Chemicals & Spirit | **Containers & Bottles**

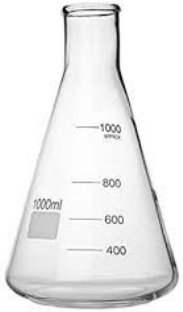
Timer | **Temperature Controller** | **Induction Stove (Option)**

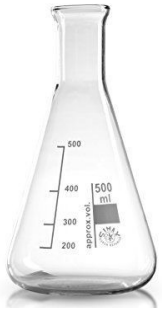


For PDA Culture (Potato Dextrose Agar)

Boiling (Test) Tubes | Petri Dish | ¼ Whisky Bottle

For Liquid Culture

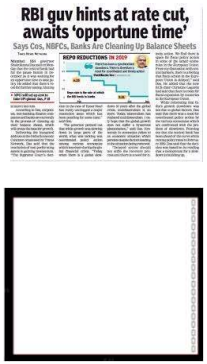




Conical Flask | Saline Bottle

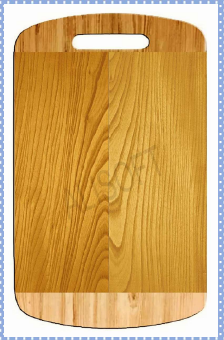
For Cultivation

Glass Bottle | Lug Jar



Cotton | Cut/ News Paper | Micropore Tape | Thread

Miscellaneous Items for the Preparation of Medium and Culture



Cutting Board **Knife** **Potato Peeler** **Sieves** **Scissor** **Spoon** **Ladle** **Spatula**

Graduated Cylinder **Measurement Jar (P.P.)** **Vessel (S.S)** **Bucket (Plastic)** **Mortar**